POLICY BRIEF · NOVEMBER 2026

From Signature to Delivery: Implementing the 2025 Nigeria–Brazil Science, Technology and Innovation Agreement

Lutfat Abimbola Usman, PhD
Hospital das Clínicas, Faculty of Medicine, University of São Paulo, Brazil
ORCID 0000-0003-1826-8614 · DOI 10.5281/zenodo.23900622

Nigeria and Brazil agreed in August 2025 to cooperate in science, technology and innovation (STI). This brief proposes how to make that agreement produce real research over the next three years, largely by connecting funding schemes and fellowships the two countries already run.

Audience: Nigeria's Federal Ministry of Innovation, Science and Technology (FMIST); Brazil's Ministry of Science, Technology and Innovation (MCTI); the foreign ministries and embassies of both countries.

Keywords: science diplomacy; South–South cooperation; research funding; researcher mobility; Nigeria; Brazil.

Executive summary

Nigeria and Brazil signed a memorandum of understanding (MoU) on STI on 25 August 2025 [1, 2]. Public accounts of the agreement list its priority areas, but say nothing about who will pay for the work, who will manage it, or how progress will be judged. Unless those questions are answered, the MoU could go the way of the two countries' Strategic Dialogue Mechanism, which met for its second session in June 2025 after a break of more than ten years [6, 7].

Much of what is needed already exists. In May 2026 Brazil's National Council for Scientific and Technological Development (CNPq) opened a R$25 million call under its ProÁfrica programme [12]. Nigeria's Tertiary Education Trust Fund (TETFund) awarded N7.5 billion to 174 research teams in the 2025 cycle of its National Research Fund [13]. Connecting these schemes would be quicker and cheaper than designing new ones.

This brief recommends five actions:

1. Establish a Joint STI Working Group with a two-country secretariat within six months.
2. Open a Nigerian co-funding window in the National Research Fund that matches ProÁfrica projects with Nigerian partners, each country funding its own researchers.
3. Fill existing fellowship routes, starting with the TWAS–FAPESP postdoctoral call (full proposals due 1 April 2027) [14].

4. Designate a science focal point in each country's embassy.
5. Report annually against a public results framework, beginning with a co-authorship baseline.

1. Context: the 2025 cooperation framework

The STI MoU was signed in Brasília by Nigeria's Minister of Innovation, Science and Technology, Uche Geoffrey Nnaji, and Brazil's Minister of Science, Technology and Innovation, Luciana Santos, with Presidents Tinubu and Lula as witnesses [1, 2]. It covers biotechnology, the bioeconomy, ocean science, energy, space, digital transformation and new materials [2].

The MoU was one of several agreements reached in 2025, and together they shape what it can realistically achieve.

Date	Instrument	Relevance to STI
25 Aug 2025	STI MoU (FMIST–MCTI) [1, 2]	Core framework. Nigeria's ministry plans to use Brazilian cotton research data with NABDA, NASRDA and the Energy Commission of Nigeria [3]
25 Aug 2025	Bilateral Air Services Agreement; direct Lagos–São Paulo service launched [1, 5]	Lowers the cost and time of researcher mobility
Jul–Aug 2025	Health ministries' MoU; Nigeria proposes a five-year joint action plan [4]	Entry point for health and pharmaceutical research
24 Jun 2025	Second session of the Strategic Dialogue Mechanism, Abuja [6]	Agreements on energy and agriculture; the Green Imperative mechanised-farming project, worth over US$1 billion, named the partnership's flagship

2. Problem statement

The two research systems differ greatly in size, and so far nothing has been set up to turn the political commitment into research on the ground.

2.1 Capacity asymmetry

Indicator	Brazil	Nigeria	African Union commitment
Gross expenditure on R&D (% of GDP)	1.1 (2020)	0.3 (2019, partial data)	At least 1.0 [10]
Researchers per million inhabitants	903 (2014)	22 (2019)	–

Source: UN Statistical Yearbook, 68th edition [9]. Reporting years differ because both countries report irregularly.

Relative to the size of its economy, Nigeria spends about a third of what the African Union commitment calls for, and it has roughly one researcher for every forty in Brazil, per head of population. This gap is the main reason the partnership is worth pursuing. It also means that joint

projects cannot assume Nigerian teams start with the same equipment and staffing as their Brazilian partners; building that capacity has to be part of the design.

2.2 Implementation gap

Four gaps stand out.

First, no body appears to be responsible for delivery. Nigeria's science minister proposed a Nigeria–Brazil Joint Innovation and Technology Working Group and a Science and Innovation Dialogue [8], but no public record could be found that either has met.

Second, routes for researcher mobility are fragile. The TWAS–CNPq programme, one of the main ways for students from developing countries to take a PhD in Brazil, did not open a call in 2026 [15].

Third, Nigeria's diplomatic missions have little dedicated science capacity. Its embassy in Washington, for instance, lists no science counsellor among its officers [19], and opportunities in host countries are easily missed as a result.

Finally, there is no agreed way to measure progress. No published baseline of Nigeria–Brazil co-authored papers could be found, so in three years' time it would be difficult to show whether the MoU has made any difference.

3. Evidence and lessons

3.1 Analytical framework

The Royal Society and the American Association for the Advancement of Science distinguish three dimensions of science diplomacy: science in diplomacy (scientific advice informing foreign policy), diplomacy for science (facilitating international research cooperation) and science for diplomacy (using scientific cooperation to improve relations) [11]. The 2025 MoU is primarily an instrument of diplomacy for science. It should therefore be judged by the research it produces and the capacity it builds, rather than by the number of documents signed.

3.2 Lessons from Brazil’s cooperation in Africa

Brazil's earlier cooperation in Africa offers two contrasting lessons.

The Cotton-4 project, run since 2009 by the Brazilian Cooperation Agency (ABC) and EMBRAPA with Benin, Burkina Faso, Chad and Mali, trained 425 African researchers, and yields on its experimental fields were reported to have roughly quadrupled [16]. It suggests that patient, long-term technical work built around training can pay off.

ProSAVANA in Mozambique shows the other side. This Brazil–Japan–Mozambique programme, inspired by an earlier Japan–Brazil programme in Brazil's Cerrado, met strong opposition from civil society over land rights, and its plans changed substantially before implementation [17]. Agricultural projects under the new MoU should therefore consult affected communities early and test whether Brazilian practices suit local conditions before scaling them up.

Some Brazilian technologies are also inexpensive to trial. Brazilian law has recognised crushed-rock soil remineralisers as an agricultural input since 2013 (Law 12.890) [20], and joint field trials on Nigerian soils would be a modest, low-risk place to start.

3.3 Instruments already in place

Instrument	Country	What it funds	Status and relevance
CNPq ProÁfrica, Call 15/2026 [12]	Brazil	R$25 million: research costs and fellowships, including for African researchers in Brazil; projects up to R$1 million over 24–36 months	Created in 2004; final decisions due by December 2026; open to partners across Africa
TETFund National Research Fund [13]	Nigeria	N7.5 billion to 174 teams in the 2025 cycle; grants of N13.7–50.0 million	No international co-funding window identified
TWAS–FAPESP postdoctoral fellowships [14]	Brazil (São Paulo State)	Up to 12 months for developing-country postdoctoral researchers at São Paulo universities	Full proposals due 1 April 2027
TWAS–CNPq PhD fellowships [15]	Brazil	Up to four years of PhD study	No call in 2026
Green Imperative [6]	Both	Over US$1 billion for mechanised farming equipment, training and service centres	Named flagship of the partnership; a natural host for applied research

3.4 Joint funding models

International joint calls commonly use one of two models [18]. In a real common pot, funders pay into a central account that distributes grants. In a virtual common pot, each national funder pays its own researchers within jointly selected projects. The virtual model suits this partnership: it avoids cross-border transfers and currency exposure, and each agency keeps its own rules. Its main weakness is that a delay on one side can stall the whole project, which synchronised calendars can reduce.

4. Policy options

Three options were considered. The third is recommended: it costs more than relying on the MoU alone, but it builds on funds that already exist and is the only one likely to leave lasting links between institutions.

Criterion	A. Rely on the MoU alone	B. Mobility first	C. Linked co-funding with governance (recommended)
Description	Declarations, ministerial visits and ad hoc contacts	Scholarships and fellowships only	Joint working group, matched funding for joint projects, mobility and annual reporting
Additional cost to Nigeria	None	Low to moderate	Moderate: about N500 million over three years for research co-funding
Time to first results	Uncertain	12–24 months	12–18 months
Lasting institutional links	Weak	Moderate, held by individuals	Strong, held by teams and institutions
Main risk	Dormancy, as with the Strategic Dialogue Mechanism	One-way flow of talent	Coordination load on two ministries and two funders

The Option C estimate assumes 10 joint projects with Nigerian co-funding of up to N50 million each, the upper end of 2025 National Research Fund grants [13]. Spread over three years, this equals about 2% of the 2025 Fund cycle per year. It excludes secretariat staff time and fellowships, which existing programmes cover. Cost, timing and durability ratings are the author's assessment.

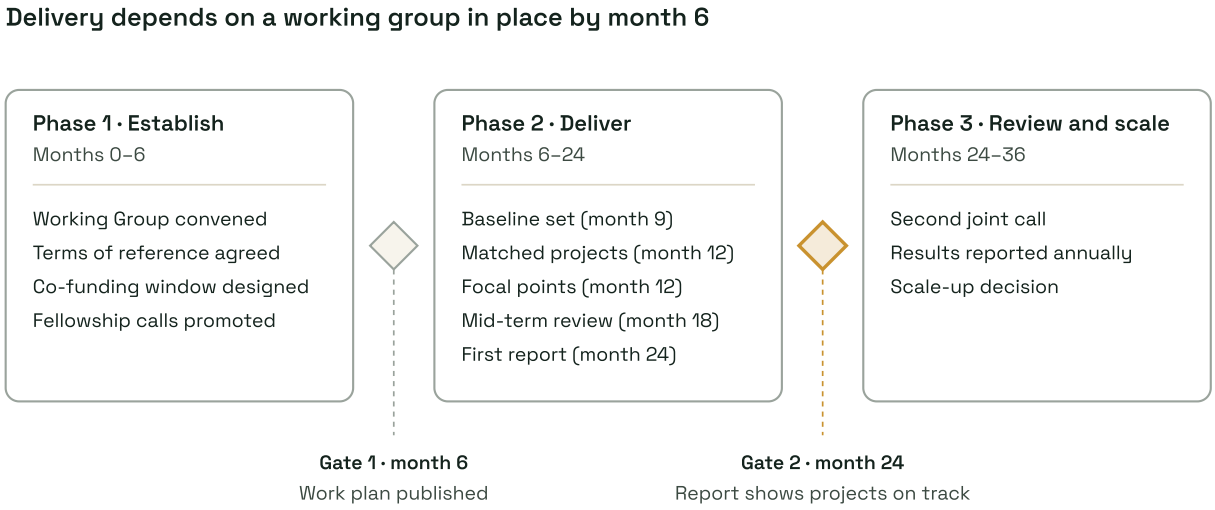
5. Recommendations

The five actions below implement Option C. Month 0 is the date both ministries adopt the plan.

#	Action	Lead institutions	Timeline	Resources
R1	**Establish a Joint STI Working Group.** One co-chair from each ministry and a secretariat of one officer per side. Publish terms of reference and a two-year work plan, and meet twice a year, alternating in person and online.	FMIST and MCTI, with both foreign ministries	Terms of reference by month 6	Existing staff time
R2	**Open a Nigerian co-funding window for joint projects.** The National Research Fund funds the Nigerian teams in ProÁfrica projects that include Nigerian partners, starting with the 2026 results. Future calls are timed together and limited to agriculture and the bioeconomy, health, and space and Earth observation. Agricultural projects include community consultation.	TETFund and FMIST, with CNPq	First matched projects by month 12	About N500 million over three years (Section 4)
R3	**Build a mobility pipeline.** Promote the TWAS–FAPESP postdoctoral call and ProÁfrica fellowships to Nigerian researchers, and ask CNPq and TWAS to reopen the PhD call in 2027. Pair each fellow with a host supervisor and a Nigerian home institution.	Nigerian Academy of Science, Nigerian Young Academy, both embassies, TWAS	Immediate; TWAS–FAPESP deadline 1 April 2027	Low: existing fellowships
R4	**Designate science focal points.** Name a science counsellor or focal point in Nigeria's embassy in Brasília and in Brazil's embassy in Abuja, supported by an advisory network of Nigerian researchers in Brazil.	Nigeria's Ministry of Foreign Affairs, FMIST, Nigerians in Diaspora Commission; Itamaraty	By month 12	One designated post per embassy
R5	**Publish an annual results report.** Report against the framework in Section 7 at each Science and Innovation Dialogue, beginning with a 2023–2025 co-authorship baseline.	Working Group secretariat	Baseline by month 9; first report by month 24	Secretariat time; open bibliometric data

6. Implementation roadmap

The plan runs over three years in three phases. Each phase should begin only after the preceding gate has been passed.



Implementation roadmap · 3 phases, 2 gates, 36 months

Gate 2 is the decision point. If the first annual report shows projects on track, the partners open a second joint call; if not, the Working Group revises the design before committing further funds.

7. Monitoring and evaluation framework

The framework follows a results chain from inputs to outcomes. Targets are proposed for month 36 and should be confirmed by the Working Group.

Level	Indicator	Baseline	Proposed target (month 36)	Data source
Input	Nigerian co-funding committed to joint projects	No dedicated window	Window established; about N500 million committed	TETFund
Activity	Working Group meetings held, with published minutes	0	At least 5	Secretariat records
Output	Joint Nigeria–Brazil projects funded	To be confirmed from ProÁfrica 2026 results	At least 10	CNPq and TETFund
Output	Researchers exchanged	Not tracked	At least 50 Nigerians hosted in Brazil and 20 Brazilians in Nigeria	Fellowship and project records
Outcome	Nigeria–Brazil co-authored publications per year	2023–2025 average, to be established	50% above baseline	OpenAlex or Web of Science
Outcome	Joint projects winning further external funding	0	At least 3	Project reports

A light mid-term review at month 18 should test whether the indicators remain realistic and adjust targets before the second joint call.

8. Risks and mitigation

The greatest risk is simply that political attention moves on. For that reason, responsibility for delivery should rest with the funding agencies as well as the ministries. The likelihood and impact ratings below are the author's judgement.

Risk	Likelihood	Impact	Mitigation
Political attention shifts after changes in leadership	Medium	High	Anchor delivery in TETFund and CNPq as well as ministries; publish annual reports
Funding released late on one side	Medium	Medium	Synchronised call calendars and a written timetable agreed between the funders
Exchange-rate volatility	High	Medium	Virtual common pot: each funder pays its own researchers in local currency
One-way flow of talent to Brazil	Medium	Medium	Joint supervision, a Nigerian home institution for every fellow, and equipment budgets on the Nigerian side
Brazilian models transferred without local fit	Medium	High	Field trials before scale-up and consultation with affected communities, drawing on the ProSAVANA experience [17]
Visa and administrative delays	Medium	Medium	Embassy science focal points to track cases and brief consular sections

9. Conclusion

The 2025 agreement is a good start, but on its own it will not produce research. Most of the pieces are already in place: Brazil's ProÁfrica call, Nigeria's National Research Fund, international fellowships and a direct flight between Lagos and São Paulo. What has been missing is someone to connect them, a modest amount of matched funding, and a public record of progress. None of this is expensive, and the first steps could be taken within the 2027 budget year.

About the author

Dr Lutfat Abimbola Usman is a Nigerian biochemist and postdoctoral researcher at Hospital das Clínicas, University of São Paulo, working on multi-omics diagnostics. She writes as a Nigerian researcher working within the Brazilian research system. She has research experience in Nigeria, Malaysia and Brazil and served as lead reviewer on an international collaborative research funding panel in 2026. The views expressed are her own and do not represent the University of São Paulo or FAPESP.

Contact: ulutfat@gmail.com · drlutfat-usman.github.io

Suggested citation: Usman, L. A. (2026). *From signature to delivery: Implementing the 2025 Nigeria–Brazil science, technology and innovation agreement*. Policy brief. São Paulo, November 2026. Zenodo. https://doi.org/10.5281/zenodo.23900622

References

1. State House Nigeria (2025, 25 August). *President Tinubu lauds Petrobras' imminent return to Nigeria as Nigeria, Brazil sign five MoUs to boost economic ties*. statehouse.gov.ng/?p=57091
2. Space in Africa (2025, 28 August). *Nigeria and Brazil to collaborate on space science and technology with strategic MoU* (source: NASRDA). spaceinafrica.com
3. Federal Ministry of Information and National Orientation (2025). *Nigeria and Brazil to strengthen bio-technology cooperation – FG*. fmino.gov.ng
4. Federal Ministry of Information and National Orientation (2025, 27 August). *Nigeria, Brazil strengthen health ties, propose five-year joint action plan*. fmino.gov.ng
5. The Guardian Nigeria (2025). *Air Peace begins Lagos–São Paulo flight after Tinubu's Brazil deal*. guardian.ng/?p=2664990
6. State House Nigeria (2025, 24 June). *At the second Dialogue Mechanism: VP Shettima lauds Nigeria–Brazil $1bn agric deal, strategic alliance*. statehouse.gov.ng
7. State House Nigeria (2025, 11 March). *Nigeria–Brazil Strategic Dialogue to resume after decade-long break*. statehouse.gov.ng/?p=53449
8. The Guardian Nigeria (n.d.). *Nigeria seeks science, tech collaboration with Brazil, Cuba through joint innovation platforms*. guardian.ng/?p=2611193
9. United Nations Statistics Division (2025). *Statistical Yearbook, 68th issue: Research and development expenditure and staff*. data.un.org
10. United Nations Economic Commission for Africa (n.d.). *Towards achieving the African Union's recommendation of expenditure of 1% of GDP on research and development*. uneca.org
11. The Royal Society and AAAS (2010). *New frontiers in science diplomacy: Navigating the changing balance of power*. London: The Royal Society. royalsociety.org
12. CNPq and MCTI (2026). *Chamada Pública CNPq/MCTI nº 15/2026 – Programa de Cooperação Afro-Brasileira em Ciência e Tecnologia (ProÁfrica)*. gov.br/cnpq
13. The Guardian Nigeria (2025). *TETFund awards N7.5b research grants to 174 teams*. guardian.ng/?p=2857911
14. TWAS (2026). *TWAS–FAPESP Postdoctoral Fellowship Programme for developing countries*. twas.org
15. TWAS (n.d.). *TWAS–CNPq Postgraduate Fellowship Programme*. twas.org
16. Revista Cultivar (2016, 18 April). *Brasil apresenta a parceiros africanos resultados de cooperação no setor algodoeiro*. revistacultivar.com.br
17. Shankland, A. and Gonçalves, E. (2016). Imagining agricultural development in South–South cooperation: The contestation and transformation of ProSAVANA. *World Development*, 81, 35–46. doi.org/10.1016/j.worlddev.2016.01.002
18. ERA-LEARN (n.d.). *Distribution and monitoring of funds*. era-learn.eu
19. Embassy of Nigeria, Washington DC (n.d.). *Mission officers*. washingtondc.foreignaffairs.gov.ng
20. Brasil (2013). *Lei nº 12.890, de 10 de dezembro de 2013*, which added soil remineralisers to the categories of agricultural inputs; as reviewed in *Revista do Regne*, UFRN. periodicos.ufrn.br